



MECH MAINT- BOILER Department
SIMHADRI

Note No :1



Ref:SMTTPP/MECH MAINT-/2023-24/LMI/DUCT/847230

Date:23/01/2024(18:44:31)

Subject:LMI for Duct & Expansion Joint Maintenance

The LMI for "Duct & Expansion Joint Maintenance" Rev-0 made in line with the new OGN COS-ISO-00-OGN/OPS/MECH/059 is being put forward for approval.

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Note No:3

A new LMI "Duct & Expansion Joint Maintenance" Rev-0 has been prepared w.r.t the new OGN COS-ISO-00-OGN/OPS/MECH/059.

Approval may kindly be accorded.

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A new LMI "Duct & Expansion Joint Maintenance" Rev-0 has been prepared w.r.t the new OGN COS-ISO00-OGN/OPS/MECH/059. Format Revised.

Approval may kindly be accorded.

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Revision -0, January 2024

LOCATION MANAGEMENT INSTRUCTION

Guidelines for Duct & Expansion Joint Maintenance

**NTPC LIMITED
SIMHADRI SUPER THERMAL POWER PROJECT**

**NTPC LIMITED
SIMHADRI SUPER THERMAL POWER PROJECT**

TECHNICAL COMPLIANCE SYSTEM

LOCATION MANAGEMENT INSTRUCTION - LMI/OGN/OPS/MECH/059

Rev.No.:0 Date: January 2024

Guidelines for Duct & Expansion Joint Maintenance

Approved for Implementation by:

HOP (NTPC-Simhadri)

Date:

Enquiries to : GM(Operation & Maintenance)

LMI APPROVAL DOCUMENT

1	TITLE OF LMI	GUIDELINES FOR DUCT & EXPANSION JOINT MAINTENANCE
2	DOCUMENT NUMBER REVISION/ ISSUE NO & DATE	LMI/OGN/OPS/MECH/059 Revision -0, Issue No.-1, January 2024
3	ORIGINAL/ REF DOCUMENT	COS-ISO-00-OGN/OPS/MECH/059 Rev. No.: 0 DEC 2023
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10	Reviewed by:	D. P. PATRA, GM (O&M)
11	Approved by	SANJAY KUMAR SINHA, HOP(NTPC-SIMHADRI)
12	DISTRIBUTION LIST	ENCLOSED.

GUIDELINES FOR DUCT & EXPANSION JOINT MAINTENANCE

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1.0 SUMMARY

1.1.	Superseded Document	NA
1.2.	Scope	This LMI applies to Boiler Flue Gas & Air Ducts from Eco outlet to Chimney inlet in all units of NTPC Simhadri (4x500MW)
1.3.	Review	Review Responsibility Boiler Maintenance Department
		Review Frequency 2 yearly basis, or as and when required, whichever is lower
		Review Approval Head of O&M will be reviewing authority and Head of Project will be approving authority for any revision in this document.

DUCT AND EXPANSION JOINT MAINTENANCE

2.0 INTRODUCTION

Ducts which are not maintained in healthy condition cause many problems like hot air leakage, dust leakage, air ingress, which is the reason of low ID Fan Margin, heat loss etc. Hence healthy ducts are required for efficient operation and clean environment and this document is prepared to address the same. The document provides general guidelines on Maintenance, various checks to be carried out before & after Overhauling or maintenance activities in Flue gas Ducts, Air Ducts including expansion joints in all PF fired boilers. The document is prepared with reference to the OGN "COS-ISO-00-OGN/OPS/MECH/059 Rev0, Dec 2023"

3.0 RECORDS OF REVISION

NIL

4.0 SAFETY ASPECTS

- 4.1 All personnel that operate, inspect, or maintain Ducts should be trained in the plant safety procedures.
- 4.2 Ducts should be isolated from all aspects before opening of Manhole doors. All Fans should be isolated, all dampers and gates in ducts should be isolated and mechanically locked.
- 4.3 Gases may be present inside duct, hence Oxygen to be checked before entry.
- 4.4 Before start of work hard barricading of all hoppers, lower-level area, start of other vertical duct from bottom portion of duct to be done.
- 4.5 Ducts horizontal plates may be eroded and may not be visible due to ash layer over it. Hence first healthiness of bottom portion to be ensured before standing over it.

- 4.6** Safety nets to be provided below ducts wherever work in bottom portion to be done. Safety nets to be provided at all locations where work is done outside of ducts and where height is more than the prescribed limit.
- 4.7** Ash may be hot hence inspection and care to be taken before cleaning.
- 4.8** All manhole doors of ducts should be opened before entering inside the duct such that more than one manhole is opened in each portion of duct.
- 4.9** 230 Volts supply lights should not be used inside the Duct, only 24 Volts supply to be used inside Ducts. RCCB to be used in all electrical supply.
- 4.10** Before cancellation of PTW, it should be ensured that all personnel and tools are removed from the Ducts.
- 4.11** All PPE's to be used.
- 4.12** Mandatory use of safety belts by people working at heights.
- 4.13** Load Test and certification of lifting tools and tackles are to be conducted by "Competent Person" as per Factories Act' 1948 and Relevant state rules. The necessary Inspection of Lifting Tackles is to be conducted by Lifting Equipment Inspector (as per COS-ISO-00- OGN/OPS/SAFT/001 Rev. No.: 0). After that only, lifting tools and tackles are to be put to use.
- 4.14** If any scaffolding need to for maintenance work then the same may be inspected by SCAFFOLDING AND LADDER INSPECTOR as per COS-ISO-00-OGN/OPS/SAFT/001 Rev. No.:0 before it is put to use.

5.0 RECORDS OF MAINTENANCE

Maintenance records provide database for future maintenance planning and generate history for understanding the erosion pattern

Maintenance records should also be maintained in electronic form and should be available on the intranet (BMD dashboard at plant webpage).

Record includes

- a) Oxygen Survey report before & after OH
- b) Duct thickness survey report
- c) Expansion joint details along with work carried out (Pre & Post OH)
(complete population)
- d) Damper and gates status along with work carried out (Pre & Post OH)
(complete population)

Checklist as provided in Appendix 1 to be maintained for each Unit Overhauling.

6.0 INSPECTION / MAINTENANCE OF Ducts

6.1 Checks before Start of Maintenance Works:

SI No	Check
1	Welding supply is available at site along with earthing strip.
2	Welding machine is Tested
3	24 V lighting supply available
4	Proper T&P and should be tested.
5	Hoist & Winch in working condition, tested and within validity dates.
6	Construction equipment like Truck, hydra, forklift for movement of material with valid LOAD TEST certificates and routes.
7	Proper approach, platform at locations where maintenance activity to be carried out.
8	Identification of leakage points/ patches in Duct
9	Pre-OH data available like Oxygen Survey report, Insulation survey report
10	Location identified for keeping Fresh material and scrap material

6.2 Inspection & Maintenance Guidelines

Following guidelines to be adhered during inspection & Maintenance activities.

- I. All duct Plates (including diversion plates) found below 50 % of original thickness to be replaced.
- II. MEJ are to be replaced if thickness is found below 50 % of original thickness (MEJ thickness to be taken at corners and bottom portion) or any crack observed in it. The average life span of MEJ is 10 years for the purpose of material planning. After 10 years in service MEJ thickness survey to be done in every OH after taking out the protection plate and to be replaced if any are found with thickness reduction by > 50%. No part replacement of MEJ is permissible it is to be replaced if any damage found after 10 years of life.
- III. All NMEJ to be replaced every 6 years and Hot air line NMEJ in every 2 years.

- IV. Refractory impregnated with wire mesh to be applied at high erosion prone area.
- V. In very high erosion areas wear plates to be used first on the base plate, onto which the plastic refractory protection to be applied.
- VI. Half cut old boiler tubes to be fitted in leading edge of diversion plates.
- VII. Full welding of Gusset plates to be done
- VIII. Bracing to Gusset plate full welding to be done
- IX. Replacement of Bracings to be done with bracings of matching size (oversize bracings should not be used).
- X. Protection angle in the direction of flue gas to be provided on bracing pipe and protection angle stitch welding to be done with bracing pipe.
- XI. Wherever holes found in duct or plates is below 50 % only grafting to be done after removal old worn-out plates.
- XII. Gate and damper seals to be checked and replacement to be done based on condition during overhauling/ opportunity.
- XIII. Insulation to be done as per Insulation survey and at the locations where insulation was removed for maintenance activities.
- XIV. Non-metallic expansion joints in cold and hot PA duct & hot Secondary air duct, wherever provided, should be replaced with Metallic expansion joints in a phased manner. A rolling plan must be prepared accordingly. While converting NMEJs to MEJ, the corrugations should not protrude in flow path.
- XV. Restoration of Guide vanes with vanes of matching profile at originally installed locations.

6.3 General Guidelines

- I. Maintenance of ducts, dampers and expansion joints shall be the responsibility of the Engineer looking after Ducts. However respective area engineers as given below shall ensure that all defects/ liabilities are rectified before giving clearance for box up
 - a. APH inlet & outlet – APH area Engr
 - b. ESP inlet and outlet – ESP area Engr
 - c. Mill individual inlet ducts – Mill area Engr

- d. PA and FD outlet – Fan area Engr
- e. Windbox – PP area Engr
- II. If CAVT is done in ducts, record of same to be kept.
- III. In case of ash accumulation in ESP inlet in ducts at partial load, Units may be run with 1 pass closed at partial load to avoid ash accumulation.
- IV. In case heavy ash accumulation is observed inside ducts, CFD analysis to be planned.
- V. Fitting of Sonic blower to avoid ash accumulation to be planned.
- VI. All Gates & dampers to be operated on opportunity to avoid jamming.
- VII. All Expansion Bellows (MEJ and NMEJs) to be covered with protection plates welded only at upstream side of flow, keeping the other end free from any restraint for expansion during operation.

7.0 APPENDIX - 1: Checklist

Sl No	Component	Status (Y/N)
1	Pre-OH Duct Survey done and report avl	
2	Pre-OH Oxygen survey avl	
3	Pre-OH Insulation Survey avl	
4	Duct Thickness Survey done & report avl	
5	Expansion joint report (Pre & Post OH) and work done record avl	
6	Damper and gates Work done report avl	
7	Application of refractory location record avl	